

C-Methylation of Phenols, Tyrosine Derivatives, and a Tyrosine Containing Peptide

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Received 15 December 1998; revised 9 February 1999; accepted 10 February 1999

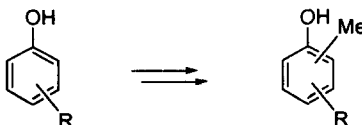
Abstract: A two step procedure is reported for the efficient C-methylation of phenolic compounds using a Stille reaction. This procedure requires no phenol protection and is tolerant to a wide variety of functional groups. © 1999 Elsevier Science Ltd. All rights reserved.

Keywords: Stille reaction, methylation, phenols.

Methods that allow easy and general access to C-methylated phenols are of interest due to the prominence of these derivatives in various natural products [1] and pharmaceuticals [2]. Ongoing investigations of quinone methides for various bioalkylating processes in our laboratories called for an efficient means for the C-methylation of numerous phenols [3]. We report a two step procedure that allows the efficient synthesis of *ortho* and *para* methylated phenol derivatives. This procedure requires no phenol protection and is tolerant to a variety of other functional groups.

Several procedures have been reported for the C-methylation of phenols (Scheme 1). A majority of these reports use the reduction of benzylic alcohols [4], aldehydes [5], benzonitriles [6], and Mannich bases [7] to produce the desired methyl groups. These procedures are limited

Scheme 1.



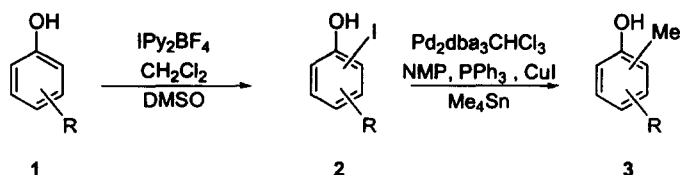
to phenols without competing reducible functionality. Directed *ortho* metalation has also been utilized for the C-methylation of phenols [8]. Phenols with base sensitive functionality are precluded from use under these conditions. Macdonald and coworkers have reported a procedure for the direct *ortho* methylation of unprotected phenols using a modified Simmons-Smith reagent [9]. They found this procedure to be incompatible with electron deficient substrates. Metal oxide catalyzed reactions of phenols and methanol under high temperature and pressure has also been reported to produce C-methylated products [10]. This method is limited to relatively simple, volatile phenolic substrates.

Methylations have also been performed on protected phenolic substrates using a Stille [11] or Negishi [12] coupling reaction. These approaches have proven to be more tolerant to a broad array of functional groups. We report an extension of the co-catalytic, palladium-copper Stille reaction [13] to convert various phenols including tyrosine, tyrosine analogs, and tyrosine containing peptides to their corresponding C-methylated derivatives. This method has proven effective in the presence of other redox active functional groups and without the need for protection of the phenol.

Phenols **1a-f** (Table 1) were converted to the desired methylated analogs by a two step procedure (Scheme 2). The phenols were first iodinated using Barluenga's reagent (IPy_2BF_4) to afford triiodo- or diiodophenols **2a-f** in excellent yields [14]. The iodophenols were converted to the methylated derivatives **3a-f** [17] via a co-catalytic, palladium-copper Stille reaction [13]. Entries 1 and 2 are examples of phenols carrying electron withdrawing, reducible functional groups. Substrates **1a** and **1b** were cleanly iodinated and efficiently converted to the corresponding trimethylphenols **3a** and **3b** in 81% and 88% overall yields, respectively. The C-methylation of phthaloyl-protected tyramine **1c** (entry 3) showed no sign of competitive iodination and was cleanly dimethylated to afford **3c** in 76% overall yield. Phthaloyl-protected octopamine **1d** (entry 4) was efficiently iodinated and dimethylated with no evidence of 1,6-elimination of the benzylic alcohol detected under the reaction conditions. This provided **3d** in 87% overall yield. Protected tyrosine **1e** (entry 5) was cleanly converted to the 3,5-dimethyl analog **3e** by the two step procedure in 73% overall yield. This approach simplifies access to the known 3,5-dimethyltyrosine derivative [15].

To examine this two step methylation in the context of a peptide derivative, we chose Fmoc-protected, truncated enkephalin **1f** (entry 6). Both the iodination and methylation of **1f** were accomplished with no sign of side reactions to afford **3f** in 75% overall yield. To our knowledge this is the first example of a tyrosine C-methylation in a peptide.

Scheme 2.

**Table 1. Phenols, Iodinated Intermediates, and Methylated Products.**

entry	compound	starting phenol 1	iodophenol 2	% yield 2	methylated phenol 3 ^b	% yield 3
1	a			98		83
2	b			96 ^a		92
3	c			97		78
4	d			95		92
5	e			99		74
6	f			93		81

^a BMTA ICl₂ was used for iodination [16].^b All products have been fully characterized [17].

A representative example of the palladium-copper, co-catalytic methylation of an iodophenol is given by the conversion of **2d** to **3d**. Iodophenol **2d** (500 mg, 934 μmol) was added to a high-pressure reaction tube containing N-methylpyrrolidinone (1.5 ml). Pd₂dba₃CHCl₃ (27 mg, 26 μmol) and triphenylphosphine (50 mg, 191 μmol) were added to the stirring solution and gently heated to ~50 °C for 10 min. Copper(I) iodide (17 mg, 91 μmol) was added to the stirring solution and again heated to ~50 °C for 10 min. After cooling to room temperature, tetramethyl tin (285 μL , 2.06 mmol) was added neat to the stirring solution. The tube was

sealed and heated to 65 °C overnight with stirring. Aqueous workup and ethyl acetate extraction followed by drying (MgSO₄) and concentration afforded the crude product. Flash chromatography provided **3d** (268 mg, 861 μmol) in 92% yield [17].

- (a) Gunatilaka, A. A. L., In *The Alkaloids*: Cordell, G. A., Ed.; Academic Press: New York, 1998; Vol. 52, pp. 1-101. (b) Gunatilaka, A. A. L., In *Progress in the Chemistry of Organic Natural Products*: Herz, W.; Kirby, G. W.; Moore, R. E.; Steglich, W. and Tamm, C., Eds.; Springer: Wien, 1996; Vol. 67, pp. 1-165. (c) Chakraborty, D. P.; Roy, S., In *Progress in the Chemistry of Organic Natural Products*: Herz, W.; Steglich, W.; Kirby, G. W.; Tamm, C., Eds.; Springer: Wien, 1991; Vol. 57, pp. 71-152.
- For recent examples, see: (a) Viracept™: Kaldor, S. W.; Kalish, V. J.; Davies II, J. F.; Shetty, B. V.; Fritz, J. E.; Appelt, K.; Burgess, J. A.; Campanale, K. M.; Chirgadze, N. Y.; Clawson, D. K.; Dressman, B. A.; Hatch, S. D.; Khalil, D. A.; Kosa, M. B.; Lubbehusen, P. P.; Muesing, M. A.; Patick, A. K.; Reich, S. H.; Su, K. S.; Tatlock, J. H. *J. Med. Chem.* **1997**, *40*, 3979-3985. (b) CellCept™: Cheng, X.-M. *Annu. Rep. Med. Chem.* **1995**, *31*, 337-355. (c) Rezulin™: Yoshioka, T.; Fujita, T.; Kanai, T.; Aizawa, Y.; Kurumada, T.; Hasegawa, K.; Horikoshi, H. *J. Med. Chem.* **1989**, *32*, 421-428.
- Hudgens, T. L.; Turnbull, K. D., *Division of Organic Chemistry*; 213th National Meeting of the American Chemical Society, San Francisco, CA, Apr. 13-17, 1997; Abs. No. 309.
- Baik, W.; Lee, H. J.; Koo, S.; Kim, B. H. *Tetrahedron Lett.* **1998**, *39*, 8125-8128.
- Mitchell, R. H.; Lai, Y.-H. *Tetrahedron Lett.* **1980**, *21*, 2637-2638.
- (a) Ram, S.; Ehrenkauf, R. E. *Synthesis* **1988**, 91-95. (b) Brown, G. R.; Foubister, A. J. *Synthesis* **1982**, 1036-1037.
- Caldwell, W. T.; Thompson, T. R. *J. Am. Chem. Soc.* **1939**, *61*, 2354-2357.
- Snieckus, V. *Chem. Rev.* **1990**, *90*, 879-933.
- Lehnert, E. K.; Sawyer, J. S.; Macdonald, T. L. *Tetrahedron Lett.* **1989**, *30*, 5215-5218.
- Kotanigawa, T. *Bull. Chem. Soc. Jpn.* **1974**, *47*, 950-953.
- For reviews see: (a) Farina, V.; Krishnamurthy, V.; Scott, W. J., *Org. React.* **1997**, *50*, 1-633. (b) Mitchell, T. N. *Synthesis* **1992**, 803-815. (c) Stille, J. K. *Angew. Chem. Int. Ed. Engl.* **1986**, *25*, 508-524.
- For reviews see: (a) Knochel, P.; Singer, R. D. *Chem. Rev.* **1993**, *93*, 2117-2188. (b) Erdik, E. *Tetrahedron* **1992**, *48*, 9577-9648. (d) Negishi, E.-I. *Acc. Chem. Res.* **1982**, *15*, 340-348.
- Farina, V.; Kapadia, S.; Krishnan, B.; Wang, C.; Liebeskind, L. S. *J. Org. Chem.* **1994**, *59*, 5905-5911.
- (a) Barluenga, J.; Garcia-Martín, M. A.; González, J. M.; Clapés, P.; Valencia, G. *Chem. Commun.* **1996**, 1505-1506. (b) Arsequell, G.; Espuña, G.; Valencia, G.; Barluenga, J.; Carlón, R. P.; González, J. M. *Tetrahedron Lett.* **1998**, *39*, 7393-7396.
- Block, P. Jr.; Coy, D. H. *J. Chem. Soc., Perkin Trans. 1* **1972**, *63*, 633-634.
- Kajigaeshi, S.; Kakinami, T.; Yamasaki, H.; Fujisaki, S.; Kondo, M.; Okamoto, T. *Chem. Lett.* **1987**, 2109-2112.
- 3a**: ¹H NMR (270 MHz, acetone-d₆) δ 10.52 (s, 1H), 7.46 (s, 1H), 6.89 (s, 1H), 2.47 (s, 3H), 2.45 (s, 3H), 2.26 (s, 3H); ¹³C NMR (68 MHz, acetone-d₆) δ 193.4, 151.8, 131.9, 131.7, 131.2, 130.2, 126.8, 18.9, 16.3, 11.1; MS (EI) *m/z* (relative intensity) 164 (M⁺, 100), 163 (91), 135 (49), 121 (18), 107 (8), 91 (34), 77 (17). **3b**: ¹H NMR (270 MHz, CDCl₃) δ 6.88 (s, 1H), 2.40 (s, 3H), 2.39 (s, 3H), 2.24 (s, 3H); ¹³C NMR (68 MHz, CD₃OD/CDCl₃, (1:1)) δ 151.2, 133.0, 130.7, 129.8, 128.2, 117.8, 110.8, 19.4, 16.4, 14.4; MS (EI) *m/z* (relative intensity) 161 (M⁺, 100), 146 (98), 118 (10), 116 (12), 91 (10), 77 (10). **3c**: ¹H NMR (270 MHz, DMSO-d₆) δ 8.02 (s, 1H), 7.84-7.77 (m, 4H), 6.70 (s, 2H), 3.70 (app. t, J = 7.7 Hz, 2H), 2.70 (app. t, J = 7.7 Hz, 2H), 2.07 (s, 6H); ¹³C NMR (68 MHz, DMSO-d₆) δ 168.2, 152.2, 134.9, 132.1, 128.94, 128.88, 124.7, 123.5, 39.7, 33.5, 17.1; MS (EI) *m/z* (relative intensity) 295 (M⁺, 23), 160 (12), 148 (100), 135 (77), 91 (16), 77 (14). **3d**: ¹H NMR (270 MHz, CD₃OD/acetone-d₆, (1:7)) δ 7.80-7.77 (m, 4H), 6.97 (s, 2H), 4.85 (dd, J = 9.1, 4.3 Hz, 1H), 3.91 (dd, J = 13.7, 9.1 Hz, 1H), 3.65 (dd, J = 13.7, 4.3 Hz, 1H), 2.16 (s, 6H); ¹³C NMR (68 MHz, CD₃OD/acetone-d₆, (1:7)) δ 169.0, 153.5, 134.9, 133.8, 133.0, 127.0, 124.6, 123.7, 71.2, 46.5, 16.7; MS (EI) *m/z* (relative intensity) 311 (M⁺, 6), 295 (13), 160 (17), 151 (100), 148 (72), 135 (54), 77 (23). **3e**: ¹H NMR (270 MHz, CD₃OD) δ 7.75-7.71 (m, 2H), 7.56-7.49 (m, 1H), 7.47-7.40 (m, 2H), 6.80 (s, 2H), 4.70 (dd, J = 8.8, 6.1 Hz, 1H), 4.16 (q, J = 7.2 Hz, 2H), 3.09 (dd, J = 13.7, 6.1 Hz, 1H), 2.95 (dd, J = 13.7, 8.8 Hz, 1H), 2.15 (s, 6H), 1.22 (t, J = 7.2 Hz, 3H); ¹³C NMR (68 MHz, CDCl₃) δ 172.4, 167.3, 152.0, 134.0, 131.9, 130.0, 128.6, 127.2, 126.7, 123.8, 61.8, 53.9, 37.2, 16.3, 14.3; MS (EI) *m/z* (relative intensity) 341 (M⁺, 4), 268 (10), 220 (26), 135 (89), 105 (100), 77 (52). **3f**: ¹H NMR (270 MHz, CD₃OD) δ 7.77 (d, J = 7.3 Hz, 2H), 7.57 (d, J = 7.3 Hz, 2H), 7.37 (t, J = 7.3 Hz, 2H), 7.31-7.22 (m, 2H), 6.81 (s, 2H), 4.37-4.11 (m, 4H), 3.95-3.76 (m, 4H), 2.98 (dd, J = 13.7, 6.3 Hz, 1H), 2.79 (dd, J = 13.7, 8.8 Hz, 1H), 2.15 (s, 6H); ¹³C NMR (75 MHz, CD₃OD) δ 172.1, 170.6, 169.1, 155.8, 150.4, 142.4, 139.8, 127.5, 126.4, 126.0, 125.4, 123.4, 122.9, 118.1, 65.4, 55.9, 47.4, 40.7, 39.0, 35.3, 14.0; HRMS-FAB (*m/z*) [M + Na]⁺ calcd for C₃₀H₃₁N₃NaO₇, 568.2054; found, 568.2049.